

Mayank Chandak

✉ mayank.chandak21@gmail.com ☎ +1 412 287 3522 📁 Portfolio 🌐 GitHub in LinkedIn

EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Aug. 2026 – May 2028

M.S. in Robotic Systems Development (MRSD)

Coursework: Advanced Computer Vision, Robot Mobility, Robot Autonomy

Indian Institute of Technology Madras, India

Jul. 2022 – May 2026

B.Tech in Mechanical Engineering — Minor in Artificial Intelligence & Machine Learning

CGPA: 9.38/10 — Ranked **Top 5/205** in Mechanical Engineering

EXPERIENCE

Undergraduate Researcher, INSPIRE Lab

May 2025 – May 2026

Surgical Phase-Aware RL for Autonomous Laparoscopic Camera Control

IIT Madras, India

- Developed a **phase-aware PPO framework for 4-DOF autonomous laparoscopic camera control**, adapting centering, zoom, and motion objectives across surgical phases.
- Built a **SOFA/LapGym simulation** with deformable tissue, articulated instruments, FEM collision modeling, and a pivotized endoscope under **RCM constraints**.
- Outperformed phase-blind RL and an analytic oracle with privileged 3D geometry, achieving **40% higher centering** and **67% lower motion jitter**.

Young Research Fellow, Computational Imaging Lab

Jul 2024 – Mar 2025

Video Restoration and Gaussian Splatting Research

IIT Madras, India

- Developed **FlareGS**, combining physics-based flare synthesis, depth-aware restoration, and **4D Gaussian Splatting** for temporally consistent urban-scene reconstruction.
- Developed **VidFlareNet** using spatial transformers, Mamba temporal modules, and depth-guided attention, outperforming Restormer and UFormer with **20.03 PSNR vs. 17.89**.

Machine Learning Engineering Intern, Quantiphi

Jun 2024 – Jul 2024

Generative AI and Document Intelligence

Mumbai, India

- Developed **Python-based LLM/GenAI solutions** and researched OCR pipelines for tabular document extraction and production-oriented AI workflows.

PUBLICATIONS

M. Chandak, et al., "FlareGS: 4D Flare Removal using Gaussian Splatting for Urban Scenes." **ICCV 2025 Workshop (2COOOL)**.

PROJECTS

MyoFootball: Reward-Driven Musculoskeletal Control for Goal-Oriented Locomotion

Demo

- Extended **MyoSuite's musculoskeletal MyoLeg environment** with ball dynamics and contact mechanics; designed a multi-stage reward pipeline for locomotion, kicking, and balance.
- Achieved **80% goal-scoring accuracy** across randomized ball placements through **100+ reward-design iterations**.

TD3 Reinforcement Learning for Robotic Manipulation in Robosuite

- Implemented **TD3 from scratch in PyTorch** for a Franka Panda arm, including actor-critic networks, target networks, replay buffers, delayed policy updates, and exploration noise.

Multi-Agent Robotics and Swarm Intelligence

- Implemented **MADDPG under CTDE** and reproduced an ant-inspired MAV communication-relay swarm for decentralized coordination and robust routing.

TECHNICAL SKILLS

Programming: Python, C++, Java, MATLAB, Bash/zsh

Robotics & Simulation: ROS 2, MuJoCo, SOFA, LapGym, Robosuite, MyoSuite, CoppeliaSim

AI/ML: PyTorch, TensorFlow, Stable-Baselines3, Reinforcement Learning, Deep Learning, Computer Vision

Tools: Git, GitHub, Linux/Ubuntu, OpenCV, NumPy, Pandas, SolidWorks, L^AT_EX

HONORS & ACHIEVEMENTS

Young Research Fellow Award (~Rs. 100K), IIT Madras: Selected **1 of 17 students from 300+** for funded undergraduate research.
Top-10 Global Rank, PBDL sRGB Low-Light Enhancement Leaderboard — **Top 5%**, JEE Advanced — **Top 1%**, JEE Main